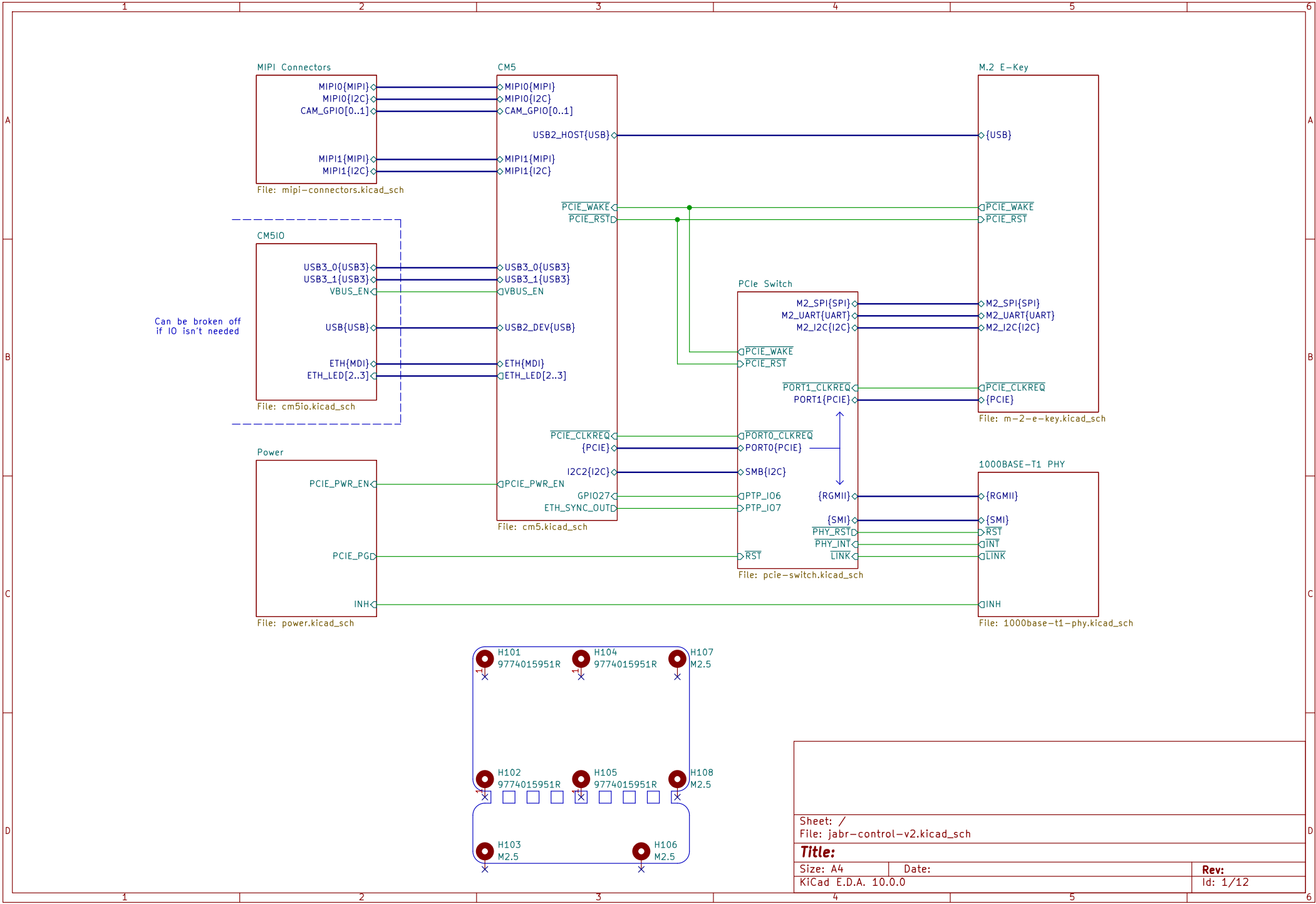




Sheet: /
File: `jabr-control-v2.kicad_sch`

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 1/12

Power In



VIN (12V-24V)

<1750mA

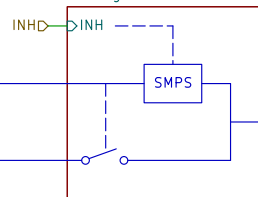
CM5IO



VBUS

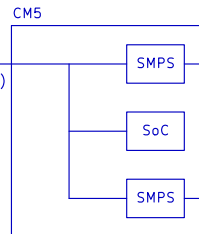
<4900mA

5V Regulator



5V

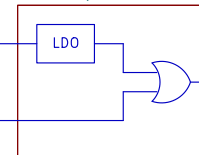
1800mA
<411mA
($\eta = 85\%$)



3V3

1V8

3V3 Always On



<77uA

<120uA

3V3_AON

<5mA

M.2 SUSCLK

6uA

CM5 VBAT

6uA

PHY Wakeup circuitry

<60uA

5V Select Circuitry

<300mA

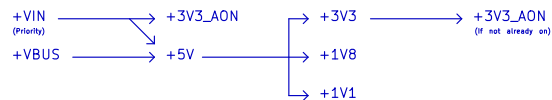
MIPI Power

The maximum power that can be consumed by the system is 13W, and the maximum power that could be consumed by peripherals (M.2 & USB-A ports) is 20.4W. Not all sources are able to provide this power so the peripheral power must be limited:

- 5V USB (3A capable): 2W for peripherals
- 5V USB (5A capable): 12W for peripherals
- 12V VIN: 5W for peripherals
- 24V VIN: 20W for peripherals

Note that the USB-A ports cannot consume more than 10W combined.

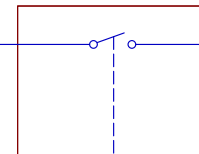
Power up sequence:



Key:

- Power
- Control

PCIe Power



1V8_PCIE

<300mA

1000BASE-T1

<100mA

PCIe Switch

20.8mA

PCIe Clock Buffer

<2721mA

($\eta = 85\%$)



3V3_PCIE

<100mA

1000BASE-T1

90mA

PCIe Switch

<3000mA

<5mA

M.2 E-Key connector

M.2 SUSCLK



1V1

<300mA

1000BASE-T1

628mA

PCIe Switch

PCIE_PWR_END PCIE_EN PCIE_PGD PCIE_PG

File: pcie-power.kicad_sch

<2185mA

USB-A Ports

Sheet: /Power/
File: power.kicad_sch

Title:

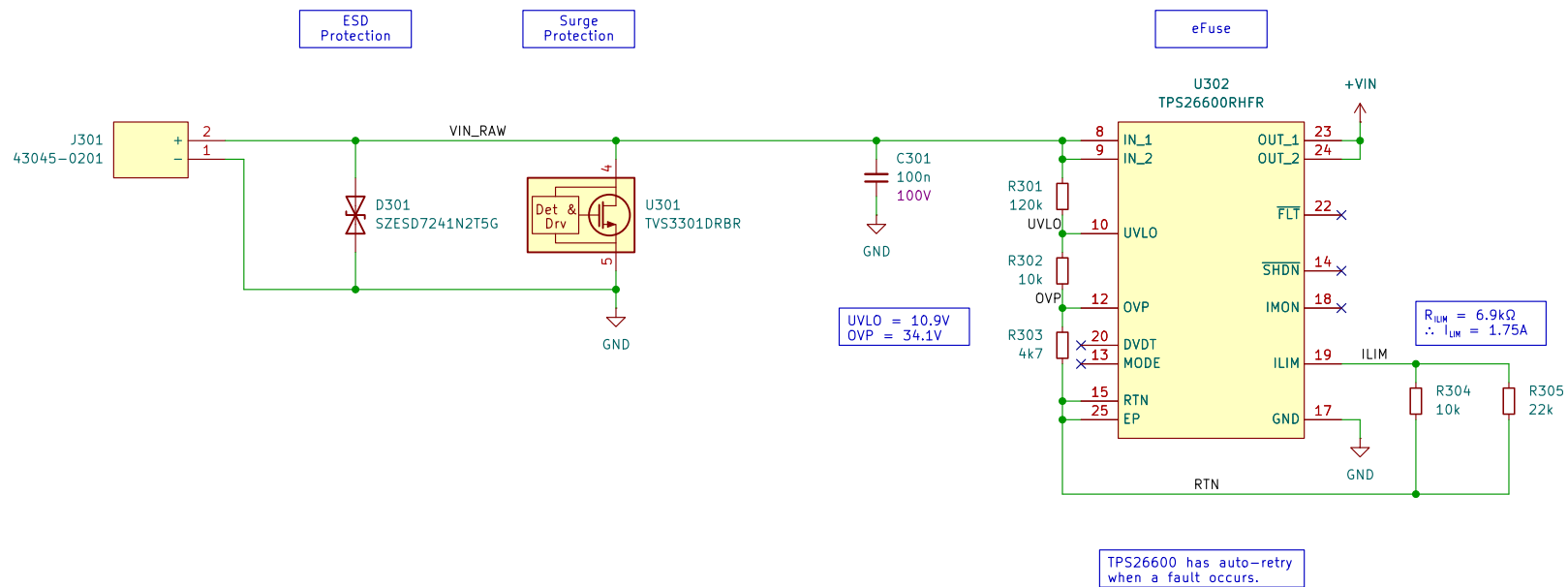
Size: A4

Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 2/12



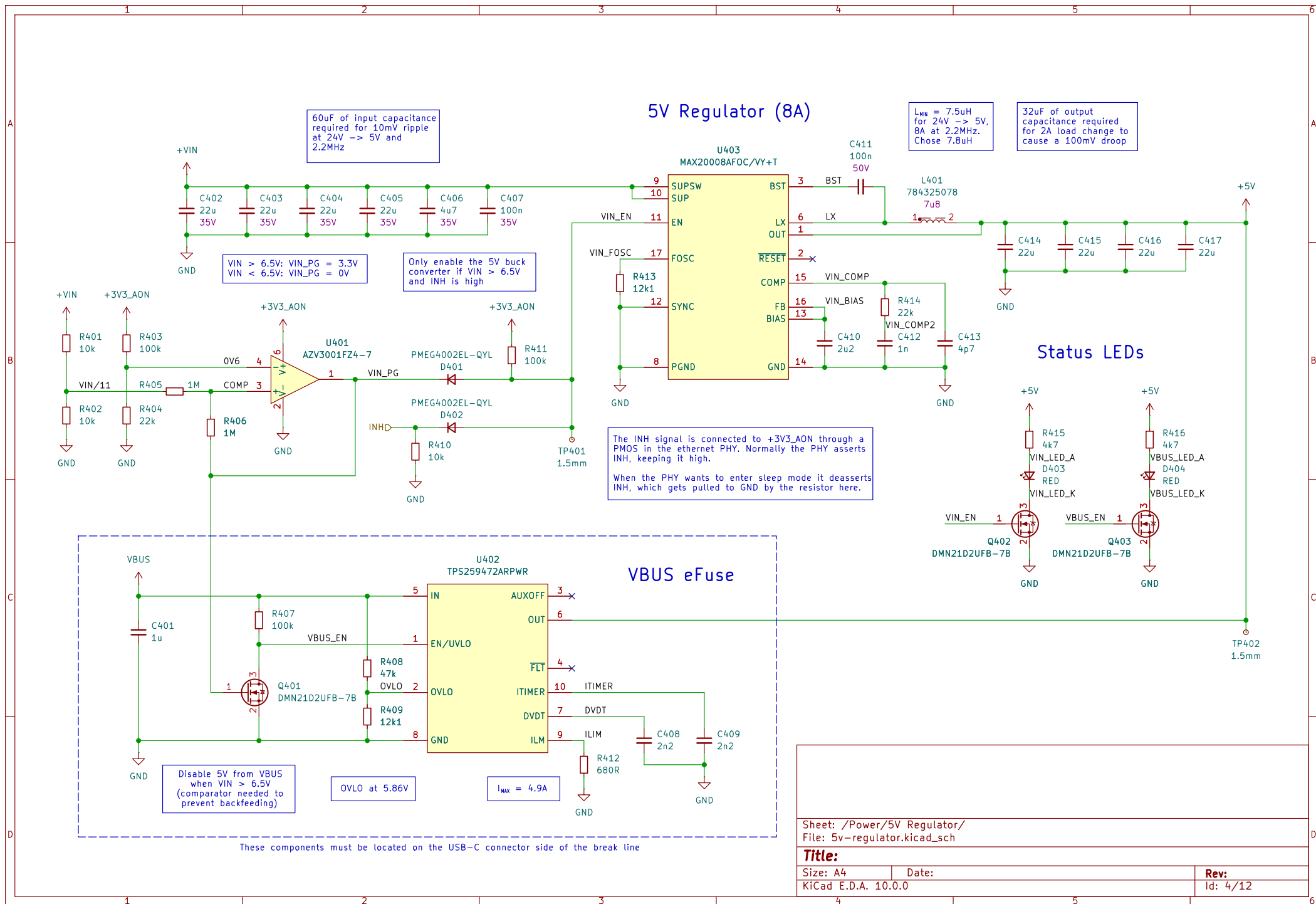
Sheet: /Power/Power In/
File: power-in.kicad_sch

Title:

Size: A4 Date:

KiCad E.D.A. 10.0.0

Rev:
Id: 3/12



Sheet: /Power/5V Regulator/
File: 5v-regulator.kicad_sch

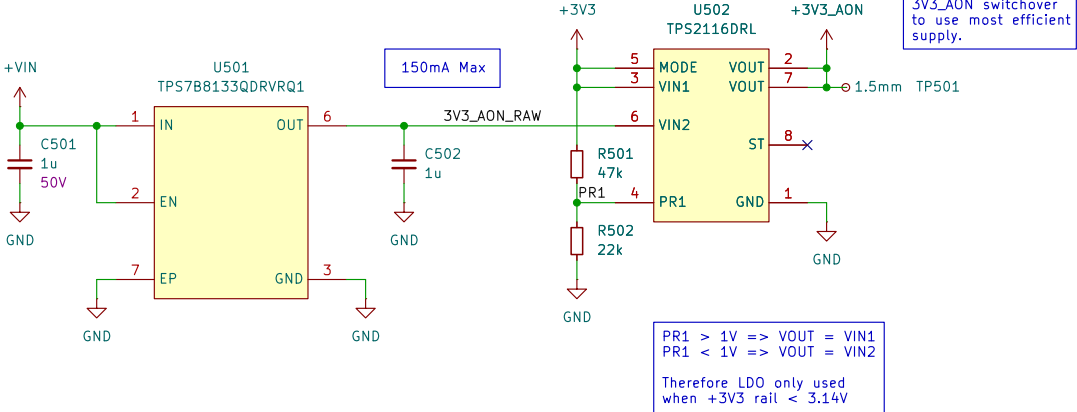
Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:
Id: 4/12

3V3 Always on supply

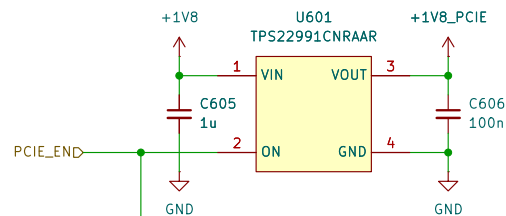


Sheet: /Power/3V3 Always On/
File: 3v3-always-on.kicad_sch

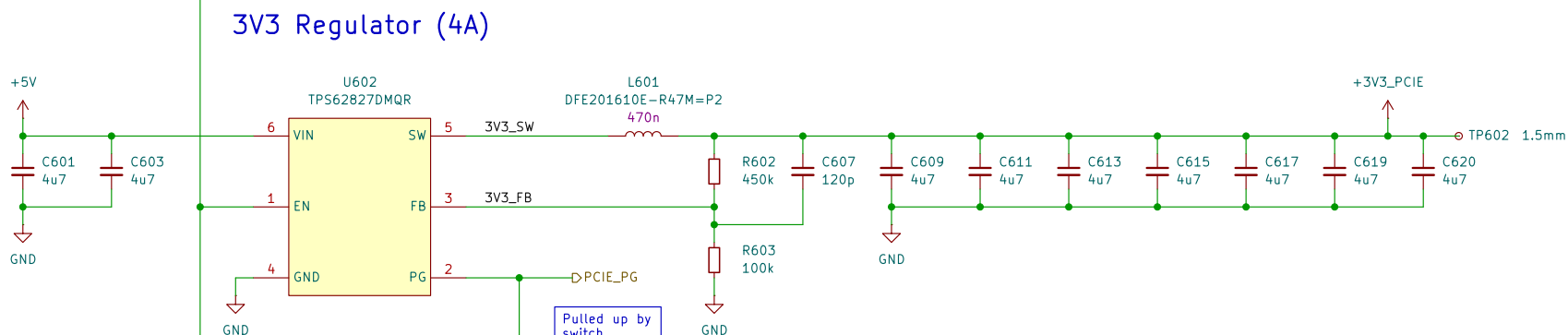
Title:

Size: A4	Date:
KiCad E.D.A. 10.0.0	

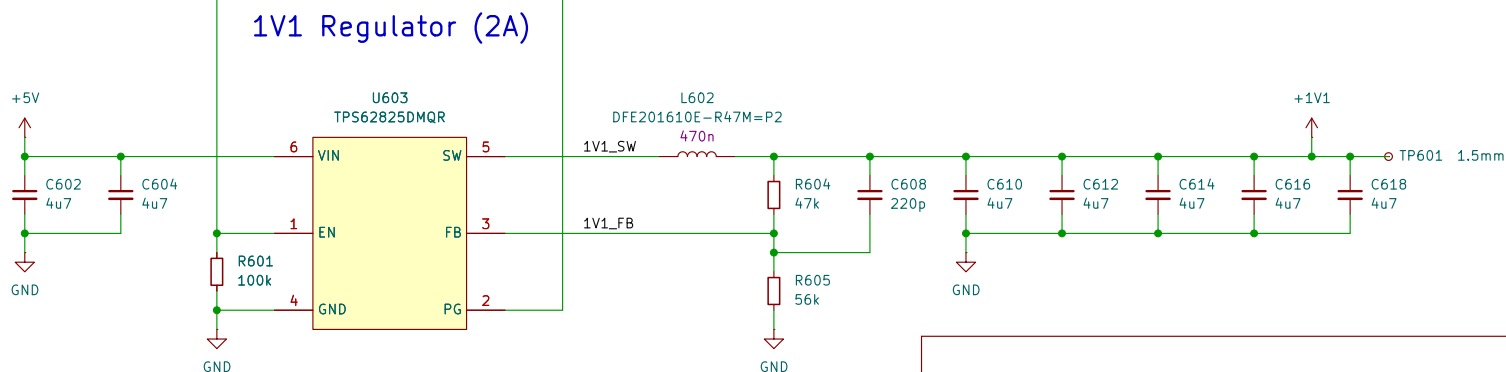
Rev:
Id: 5/12



1V8 Load Switch (3A)



3V3 Regulator (4A)



1V1 Regulator (2A)

Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

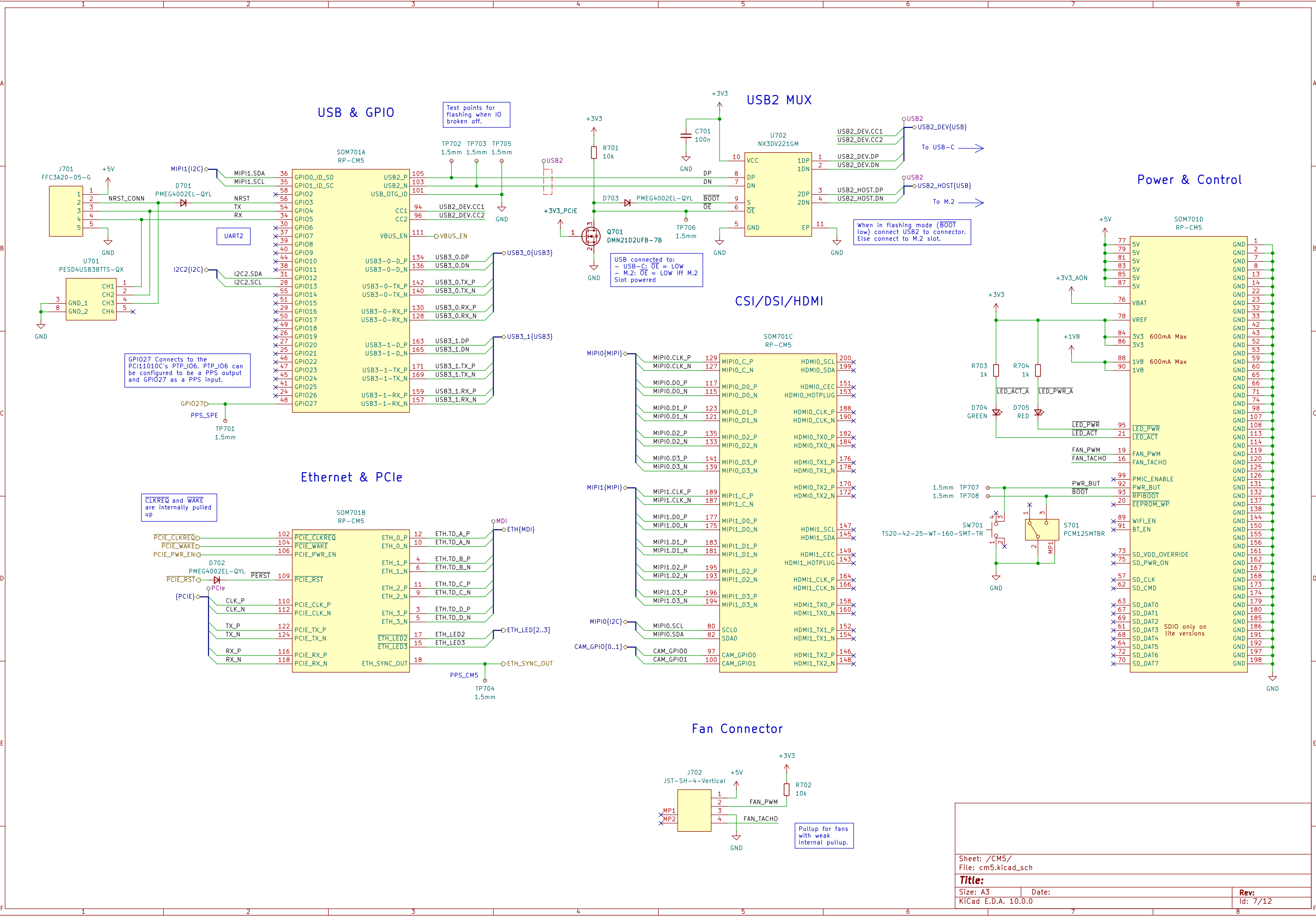
Sheet: /Power/PCIE Power/
File: pcie-power.kicad_sch

Title:

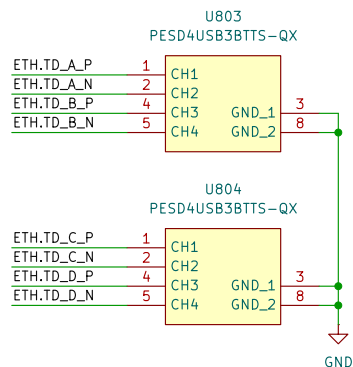
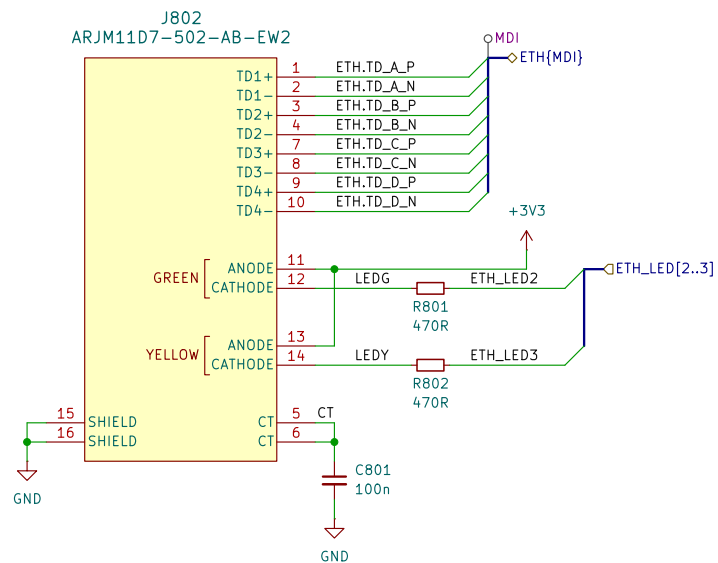
Size: A4 Date:

KiCad E.D.A. 10.0.0

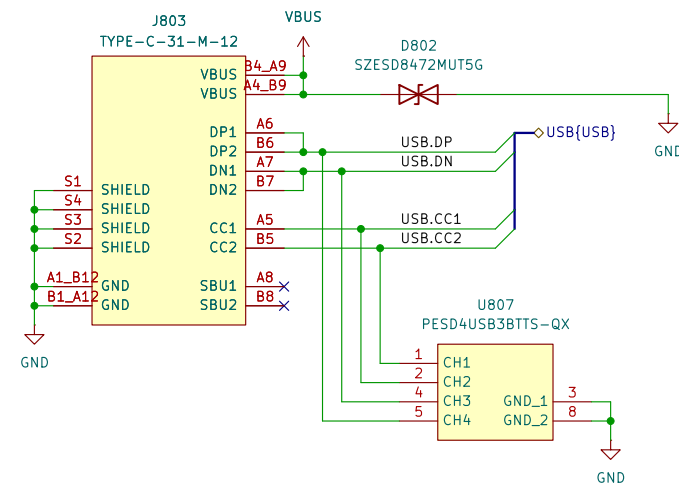
Rev:
Id: 6/12



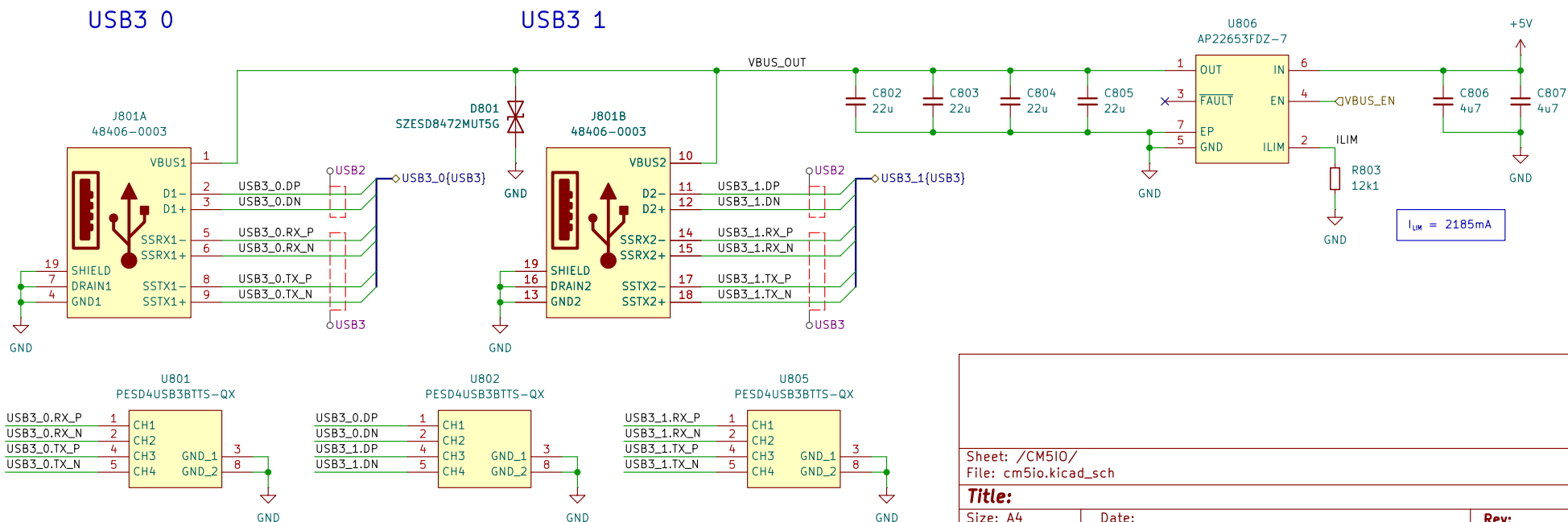
Ethernet RJ45



USB2



VBUS Load Switch



Sheet: /CM510/
File: cm510.kicad_sch

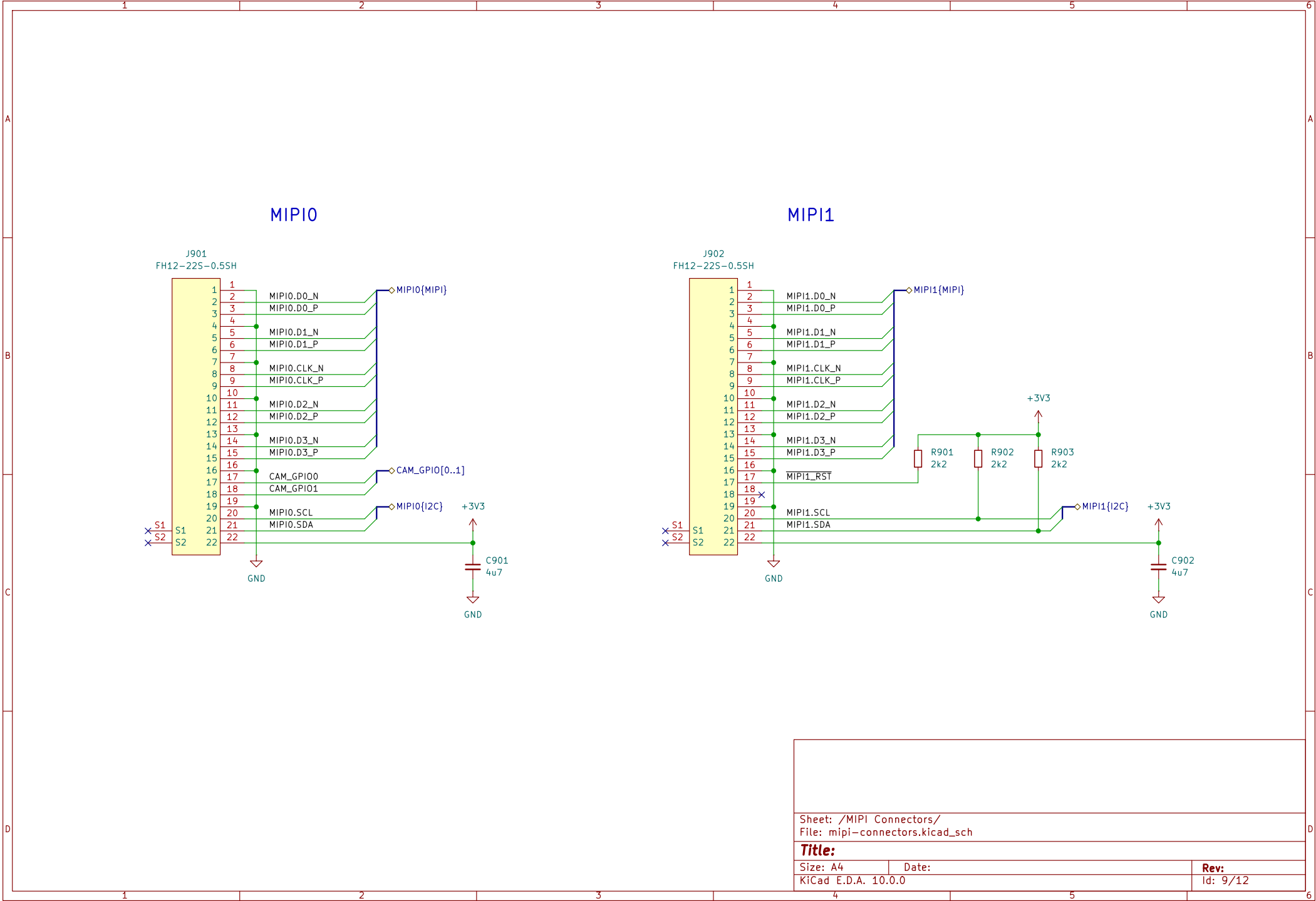
Title:

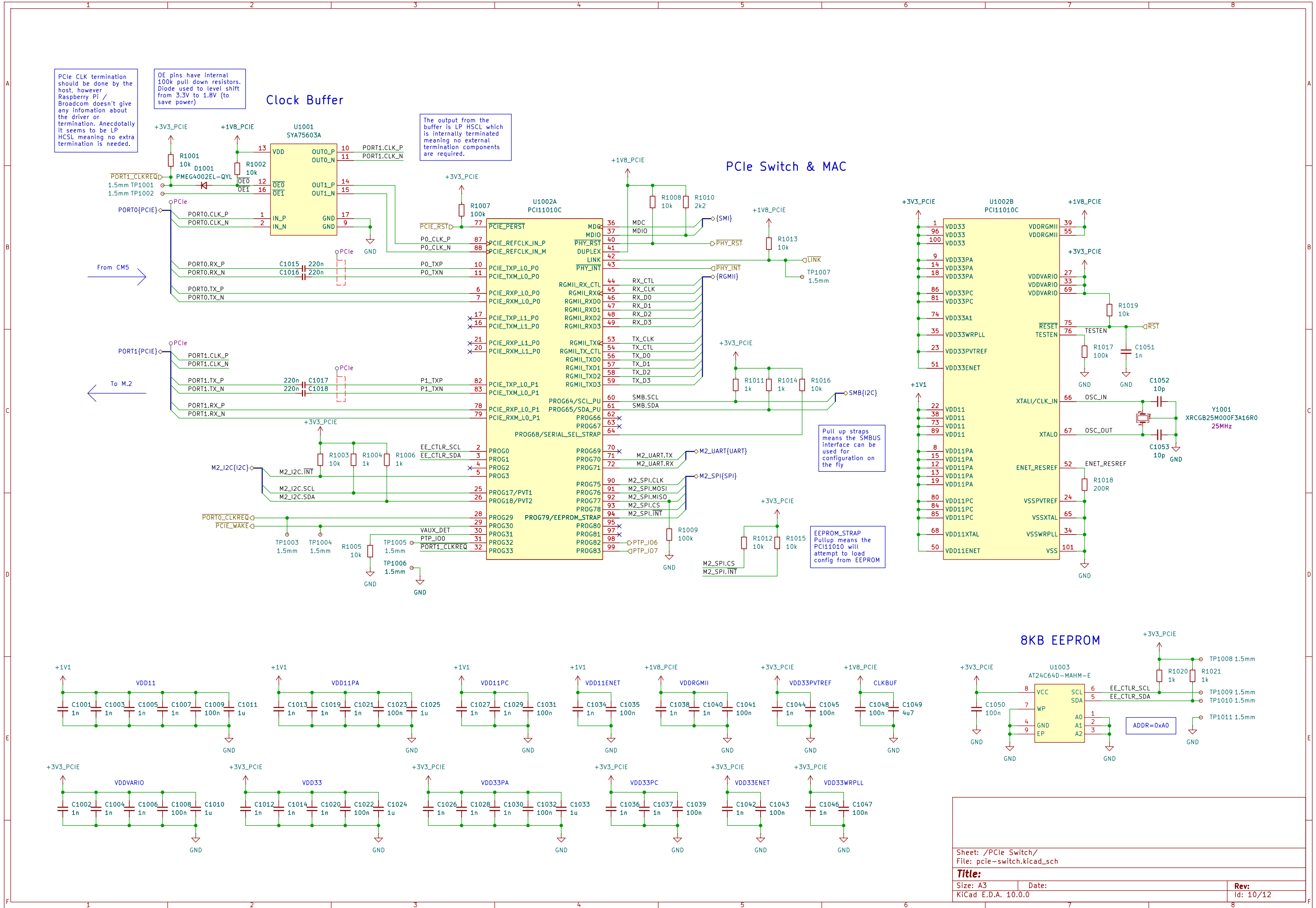
Size: A4 Date:

KiCad E.D.A. 10.0.0

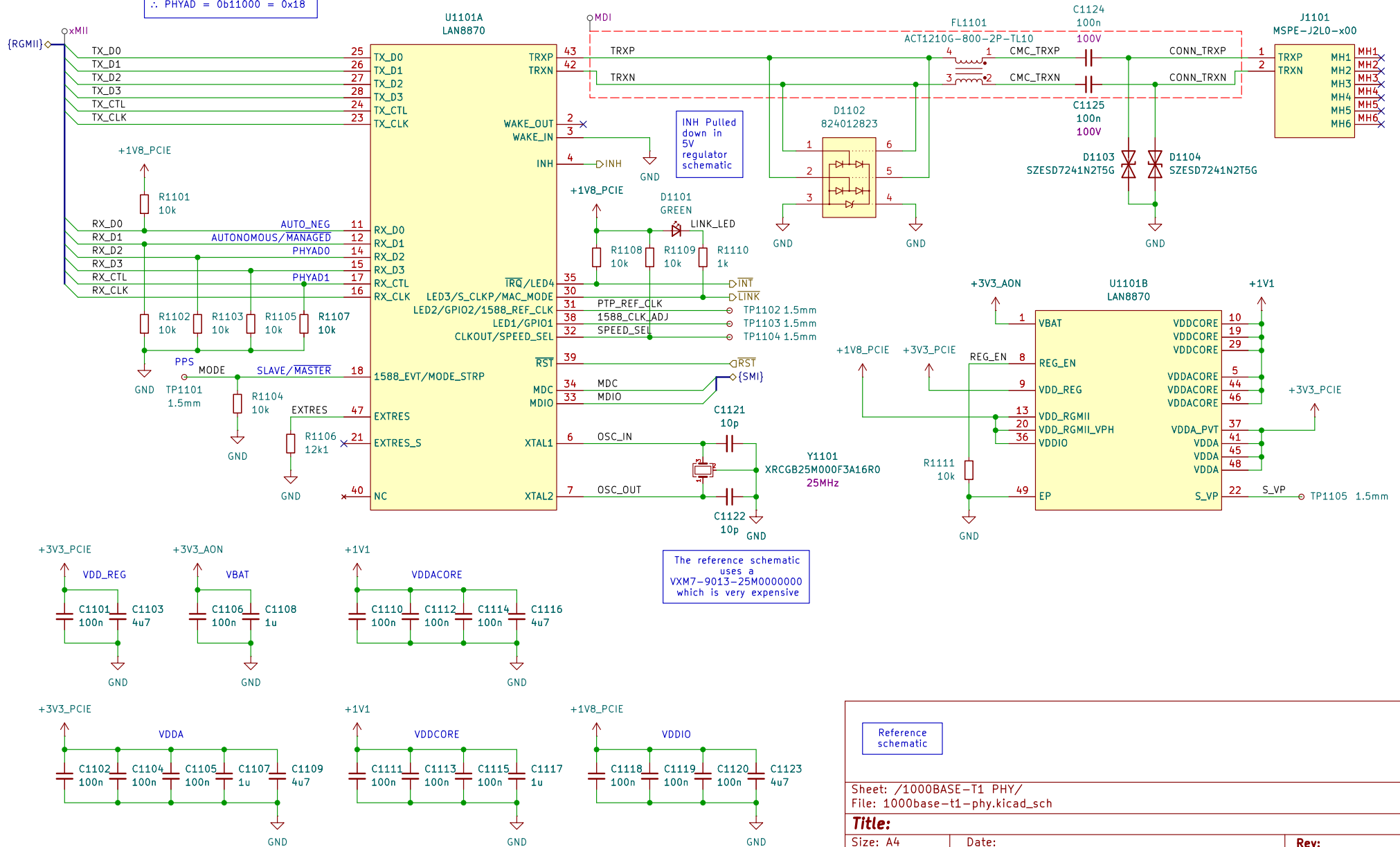
Rev:

Id: 8/12



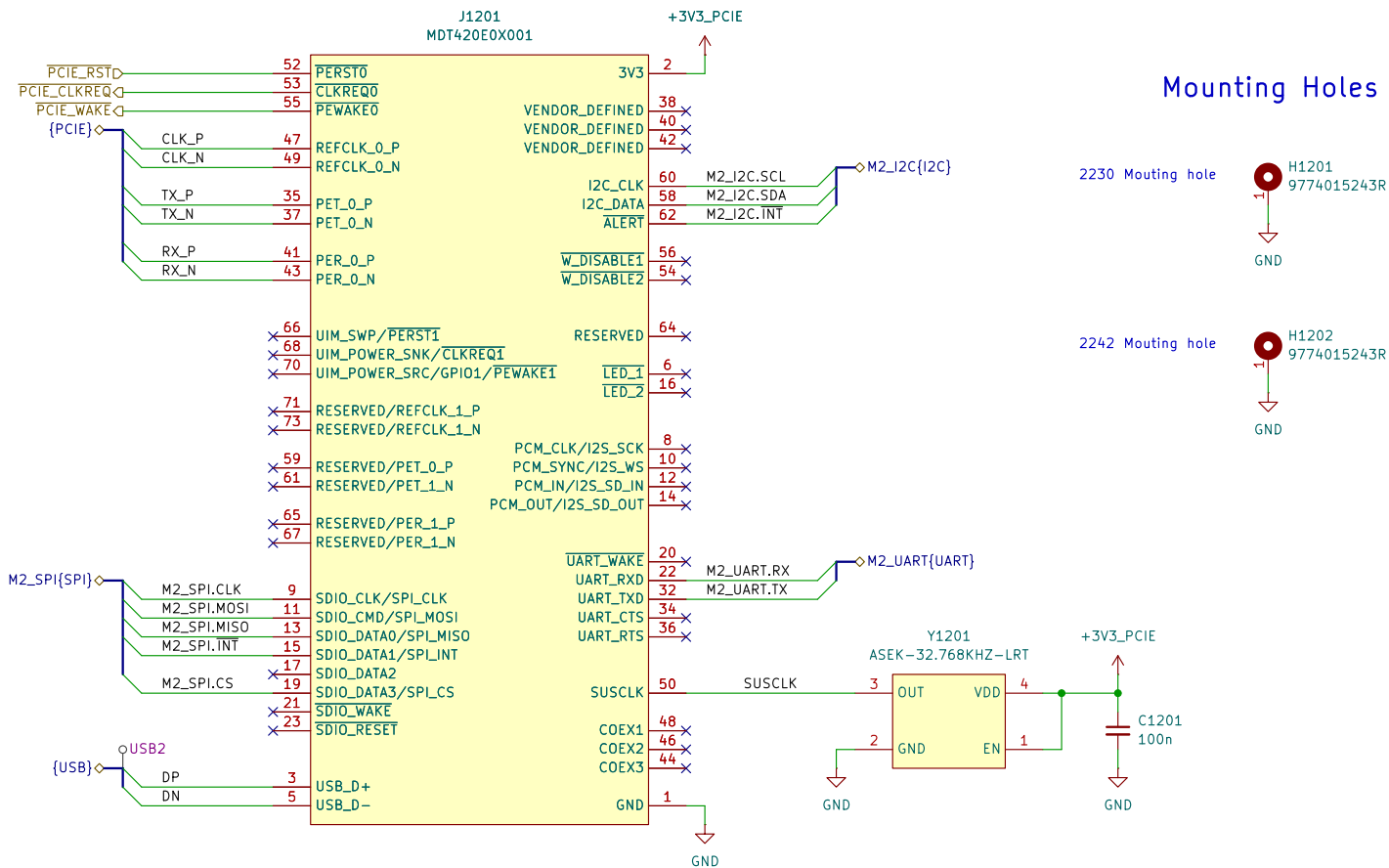


When in managed mode:
 - SPEED_SEL = PHYAD4
 - AUTO_NEG = PHYAD3
 - MODE = PHYAD2
 $\therefore$ PHYAD = 0b11000 = 0x18



Reference schematic		
Sheet: /1000BASE-T1 PHY/ File: 1000base-t1-phy.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 11/12

M.2 E-Key



Sheet: /M.2 E-Key/
File: m-2-e-key.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:

Id: 12/12